There are fast quantum algorithms for factoring enormous composite numbers into their prime factors, number theory, group theory, discrete logarithms. elliptical curve.

Peter Shor.

Quantum-resistance cryptography
(whole security infrastructure)

→ Lattice problems.

→ $200 billion bitcoins locked away.
↳ elliptical curve.

Grover's algorithm - Search ($\sqrt{n}$)

Every algorithm can be groverized, but its not exponential speedup, merely root squared speedup.

Quantum error correction works for protecting 1 Qbit, million times overhead.

So $\sqrt{n} \times 10^6 > n$ for $n < 10^6$

optimization is true $> 10^{12}$ size

Ewin Tang UW grad student

now Princeton

- Dequantizing.